

DATA 505 (Statistics Using R): Exam 2

Name: _____

2025-11-11

Instructions:

The exam will have two parts:

- **Part 1 (closed computer):** Multiple choice and short answer questions. You may not use your computer or any notes. You will write answers on this exam sheet.
- **Part 2 (open computer):** Coding and analysis problems. You may use your computer and any online resources that are "not alive" (i.e., you may freely use online resources, but you may not communicate with any other person). You will submit a completed Quarto document.
 - Once you turn in Part 1, you may open your computer and download the blank Quarto script for Part 2.
 - You may not return to Part 1 after turning it in and beginning the open-computer portion.

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TURN OVER WHEN INSTRUCTED

Part 1A (closed computer): Multiple choice

(3 points)

Which of the following correctly fits a simple linear regression model predicting y from x in R?
(Circle the correct answer)

- A. `lm(x ~ y, data = df)`
- B. `lm(y ~ x, data = df)`
- C. `regression(y, x, data = df)`
- D. `linear_model(y = x, df)`

(3 points)

In the regression equation $Y = \beta_0 + \beta_1 X + \varepsilon$, what does β_1 represent?

- A. The expected value of Y when X equals zero
- B. The change in Y for a one-unit increase in X
- C. The total variance explained by the model
- D. The correlation between X and Y

(3 points)

You fit a regression model and get an R-squared value of 0.64. What does this mean?

- A. 64% of observations fall within the confidence interval
- B. The model's predictions are correct 64% of the time
- C. 64% of the variance in the response variable is explained by the predictor(s)
- D. There is a 64% probability that the relationship is real

(3 points)

Which R function would you use to compare two nested regression models using an F-test?

- A. `summary()`
- B. `anova()`
- C. `confint()`
- D. `predict()`

(3 points)

What is the main difference between a confidence interval for the mean response and a prediction interval for a new observation?

- A. They use different statistical tests
- B. Prediction intervals are always narrower than confidence intervals
- C. Prediction intervals account for additional uncertainty about individual observations
- D. Confidence intervals cannot be calculated for regression models

(3 points)

In multiple regression with categorical predictors, what is a “reference category”?

- A. The category with the highest frequency in the data
- B. The baseline level that other categories are compared against
- C. The category that best predicts the outcome
- D. The category that R randomly selects first

(3 points)

Which of the following best describes the purpose of writing functions in R?

- A. To make code run faster
- B. To avoid repeating code and improve readability
- C. To ensure code produces graphs
- D. To automatically fix errors in data

(3 points)

Consider the following function skeleton:

```
calculate_statistic <- function(x, ...) {  
  # function body  
}
```

What does the ... (ellipsis) argument allow?

- A. It creates an infinite loop
- B. It passes additional arguments to other functions called within
- C. It's a syntax error and will not run
- D. It creates a pause in execution

(3 points)

Consider comparing two regression models: - Model 1: `mpg ~ wt` - Model 2: `mpg ~ wt + hp`

The ANOVA F-test gives a p-value of 0.0015. What can we conclude at $\alpha = 0.05$?

- A. Model 1 is significantly better than Model 2
- B. Adding `hp` does not significantly improve the model
- C. Adding `hp` significantly improves the model
- D. Neither model is valid

(3 points)

What is the primary benefit of organizing analysis code into separate function files (e.g., `01_load.R`, `02_clean.R`, `03_model.R`)?

- A. It makes the code run in parallel
- B. It improves code organization, testing, and reusability
- C. It automatically generates documentation
- D. It prevents syntax errors

Part 1B (closed computer): Short answer

Write your answers in complete sentences where appropriate.

Consider the following regression output predicting house prices (in thousands of dollars) from square footage:

Call:

```
lm(formula = price ~ sqft, data = house_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-38.15	-13.67	-1.75	11.61	65.81

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	52.141668	8.956484	5.822	7.38e-08 ***
sqft	0.147901	0.004275	34.596	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.41 on 98 degrees of freedom

Multiple R-squared: 0.9243, Adjusted R-squared: 0.9235

F-statistic: 1197 on 1 and 98 DF, p-value: < 2.2e-16

(4 points)

Write the fitted regression equation for this model (in the form: $\widehat{\text{price}} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{sqft}$).

(4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between square footage and house price?

(4 points)

Based on the p-value for the slope, is there statistically significant evidence of a linear relationship between square footage and price at the $\alpha = 0.05$ level? Justify your answer.

(4 points)

Using this model, what would you predict for the price of a house with 2500 square feet? Show your calculation.

(4 points)

Write R code that would create a scatterplot of price vs. square footage with the regression line added. Your code doesn't need to include cosmetic details like custom labels.

(3 points)

Briefly explain why extrapolating this model to predict the price of a 10,000 square foot house might be problematic.